

# **SOCIOL 723: Social Statistics II**

Week 7, Instrumental Variables

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# Today

Instrumental Variable

What Does IV Estimate?

Instrument Strength

Four Canonical Applications

Applications in Sociology

Two-Stage Least Squares

Inference Under Weak Instruments

Judge and Examiner Designs

Shift-Share Instruments

Practice

★ = essential, you must understand this

★ = for understanding, not examined

## A Change of Strategy

Week 6 assumed we had measured every confounder and worked on *how* to adjust. That assumption is usually false, and no diagnostic can rescue it.

### The instrumental variables idea

Give up on controlling for confounders. Instead, find a source of variation in the treatment that is *as good as randomly assigned*, and use only that variation.

- We throw away most of the variation in  $D$  and keep a sliver we can defend. Precision falls; credibility rises.
- The price is steep and often hidden: we no longer estimate the ATE, and the assumptions are as untestable as ignorability was, just different, and usually easier to argue about in substantive terms.

# IV

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## What Goes Wrong, in Scalar Form ★

We want  $\beta$  in  $Y_i = \alpha + \beta D_i + \epsilon_i$ , but  $D_i$  is **endogenous**,  $\text{Cov}(D_i, \epsilon_i) \neq 0$ . OLS gives

$$\begin{aligned}\hat{\beta} &= \frac{\text{Cov}_n(D_i, Y_i)}{\text{Var}_n(D_i)} = \frac{\text{Cov}_n(D_i, \alpha + \beta D_i + \epsilon_i)}{\text{Var}_n(D_i)} && \text{substitute the model for } Y_i \\ &= \beta + \frac{\text{Cov}_n(D_i, \epsilon_i)}{\text{Var}_n(D_i)} && \text{covariance is linear; the constant drops} \\ &\xrightarrow{p} \beta + \frac{\text{Cov}(D_i, \epsilon_i)}{\text{Var}(D_i)} && \text{LLN: sample moments to population moments}\end{aligned}$$

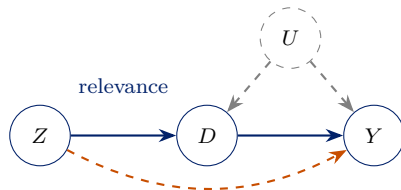
The second term does not shrink with  $n$ : OLS is biased *and* inconsistent.

**Three sources of  $\text{Cov}(D_i, \epsilon_i) \neq 0$ :** **omitted variables** (ability, family resources sit in  $\epsilon$  and move with  $D$ ; Week 1's OVB formula); **simultaneity** ( $Y$  also causes  $D$ : police and crime); **selection** into the sample on a collider (Week 5).

**A different problem, measurement error in  $D$ :** classical error in  $D$  shrinks the OLS slope toward zero, by the reliability ratio  $\text{Var}(D^*)/\text{Var}(D)$ . A known direction, unlike the three above.

## An Instrument ★

An instrument  $Z_i$  moves  $D_i$  but is otherwise unrelated to  $Y_i$ .



must not exist

## An Instrument ★ (cont.)

### Three conditions, stated carefully

1. **Relevance:**  $\text{Cov}(Z_i, D_i) \neq 0$ . **Testable.**
2. **Independence:**  $Z_i$  is as good as randomly assigned, no arrow into  $Z$  from anything that also causes  $Y$ .
3. **Exclusion:**  $Z$  affects  $Y$  *only* through  $D$ . **Not testable.**

Conditions 2 and 3 are distinct and are often conflated. A randomized lottery satisfies 2 by construction and can still violate 3.

## The Wald Estimator ★

Take covariances of  $Y_i = \alpha + \beta D_i + \epsilon_i$  with  $Z_i$ :

$$\text{Cov}(Z_i, Y_i) = \beta \text{Cov}(Z_i, D_i) + \underbrace{\text{Cov}(Z_i, \epsilon_i)}_{=0} \implies \boxed{\beta_{IV} = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, D_i)}}$$

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With a *binary* instrument this is a ratio of two differences in means:

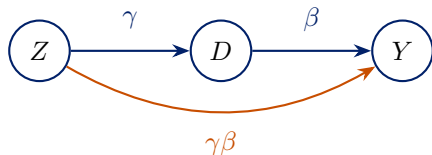
$$\beta_{IV} = \frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[D_i | Z_i = 1] - E[D_i | Z_i = 0]} = \frac{\text{reduced form}}{\text{first stage}}.$$

Read it as: *the effect of the instrument on the outcome, scaled up by how much the instrument moved the treatment.*

## Intuition for the Ratio ★

Divide numerator and denominator by  $\text{Var}(Z_i)$ :

$$\beta_{IV} = \frac{\text{Cov}(Z_i, Y_i) / \text{Var}(Z_i)}{\text{Cov}(Z_i, D_i) / \text{Var}(Z_i)} = \frac{\text{coefficient from regressing } Y \text{ on } Z}{\text{coefficient from regressing } D \text{ on } Z}.$$



- The instrument's total effect on  $Y$  travels along one path and equals  $\gamma\beta$ . The first stage recovers  $\gamma$ . Dividing isolates  $\beta$ .
- If  $\gamma$  is small, we are dividing by something close to zero. Hold that thought; it is the whole of the weak-instrument problem.

# LATE

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## Four Types of People ★

With a binary instrument and binary treatment, write  $D_i(z)$  for the treatment unit  $i$  would take if assigned  $Z_i = z$ . Every unit is one of four types:

| Type             | $D_i(0)$ | $D_i(1)$ |                             |
|------------------|----------|----------|-----------------------------|
| <b>Compliers</b> | 0        | 1        | do what the instrument says |
| Always-takers    | 1        | 1        | treated regardless          |
| Never-takers     | 0        | 0        | untreated regardless        |
| <b>Defiers</b>   | 1        | 0        | do the opposite             |

- We never observe anyone's type: we see  $D_i(1)$  or  $D_i(0)$ , never both. Type is a latent variable, and this is the same fundamental problem as Week 5, one level up.

## Four Types of People ★ (cont.)

- Only **compliers** have their treatment moved by the instrument.
- **Monotonicity** assumes there are no defiers: the instrument pushes everyone in the same direction, or not at all.

## Deriving the LATE: the Numerator

Write the reduced form using the switching equation  $Y_i = Y_i(D_i(Z_i))$  and split by type. Because  $Z$  is independent of potential outcomes and types,

$$\begin{aligned}
 & E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0] \\
 &= \Pr(\text{AT}) \underbrace{(E[Y_i(1) \mid \text{AT}] - E[Y_i(1) \mid \text{AT}])}_{=0} \\
 &+ \Pr(\text{NT}) \underbrace{(E[Y_i(0) \mid \text{NT}] - E[Y_i(0) \mid \text{NT}])}_{=0} \\
 &+ \Pr(\text{C}) (E[Y_i(1) \mid \text{C}] - E[Y_i(0) \mid \text{C}]) \\
 &- \Pr(\text{D}) (E[Y_i(1) \mid \text{D}] - E[Y_i(0) \mid \text{D}]).
 \end{aligned}$$

Always- and never-takers take the *same* treatment under both values of  $Z$ , so **by exclusion** their outcomes are unchanged and they contribute zero. **Monotonicity** kills the last line by assuming  $\Pr(\text{D}) = 0$ , note that defiers would enter with the *wrong sign*.

## Deriving the LATE: the Denominator ★

The first stage,  $E[D_i | Z_i = 1] - E[D_i | Z_i = 0]$ , recovers the complier share:

$$\begin{aligned} \text{first stage} &= \Pr(C) + \Pr(AT) - \Pr(AT) - \Pr(D) && \text{by type} \\ &= \Pr(C) - \Pr(D) \\ &= \Pr(C) && \text{no defiers} \end{aligned}$$

Dividing:

The LATE theorem (Imbens and Angrist 1994)

$$\frac{E[Y | Z = 1] - E[Y | Z = 0]}{E[D | Z = 1] - E[D | Z = 0]} = E[Y_i(1) - Y_i(0) | i \text{ is a complier}].$$

**IV does not estimate the ATE.** It estimates the average effect for the subpopulation whose treatment status the instrument actually changed, a group you cannot identify by name, defined by the instrument you happened to choose.

## Living With the LATE ★

- If effects are homogeneous,  $LATE = ATE$  and none of this matters. That is rarely a defensible assumption in sociology.
- Different instruments for the same treatment estimate *different* LATEs. When two IV papers disagree, that is not necessarily a contradiction; it may be two correct answers to two different questions.
- **You can describe the compliers even though you cannot name them:** their share is the first stage, and their covariate means follow because always-takers and never-takers are observed groups (Abadie's  $\kappa$ ).
- **Compliers are often unusual:** for quarter-of-birth they are marginal dropouts; for judge-leniency designs they are defendants near the decision margin. Whether that group is the policy-relevant one is a substantive question you must answer, not a technicality.

**Always characterize your compliers:** without that, a LATE is an average effect for a group nobody can describe.

## Describing the Compliers (Abadie 2003)

Binary  $Z$  and  $D$ , no defiers. Among  $Z = 0$  the treated are always-takers; among  $Z = 1$  the untreated are never-takers. So  $\Pr(AT)$ ,  $\Pr(NT)$ ,  $E[X | AT]$ ,  $E[X | NT]$  are observed,  $\Pr(C)$  is the first stage, and the complier mean is what is left of the population mean:

$$E[X | C] = \frac{E[X] - \Pr(AT) E[X | AT] - \Pr(NT) E[X | NT]}{\Pr(C)}$$

**Card:**  $D$  = more than one year of college,  $Z$  = near a four-year college;  
 $\Pr(AT) = 0.33$ ,  $\Pr(NT) = 0.55$ ,  $\Pr(C) = 0.12$  (figs/w7\_card.R).

|                    | everyone | always-takers | never-takers | compliers |
|--------------------|----------|---------------|--------------|-----------|
| share Black        | 0.23     | 0.15          | 0.29         | 0.21      |
| share South        | 0.40     | 0.50          | 0.38         | 0.26      |
| share metropolitan | 0.71     | 0.58          | 0.78         | 0.77      |

Compliers are less Southern and more urban than always-takers. Dividing by  $\Pr(C) = 0.12$  amplifies noise: rough numbers.

## Can You Check Monotonicity? (1) What It Implies

Not directly,  $D_i(1)$  and  $D_i(0)$  are never both seen; but “no defiers” can be *refuted*.

- **One sign everywhere.** If  $Z$  only pushes  $D$  up, then  $\Pr(D = 1 \mid Z = 1, X = x) \geq \Pr(D = 1 \mid Z = 0, X = x)$  in every subgroup; a negative first stage somewhere is a rejection. Card (figs/w7\_card.R):

|             | South | non-South | metro | non-metro | Black | other |
|-------------|-------|-----------|-------|-----------|-------|-------|
| first stage | 0.31  | 0.34      | 0.37  | 0.27      | 0.28  | 0.35  |

- **Balke–Pearl inequalities** (binary  $Z, D$ ). Switching  $Z$  on can only *add* treated units, so for any outcome set  $A$ ,  $\Pr(Y \in A, D = 1 \mid Z = 1) \geq \Pr(Y \in A, D = 1 \mid Z = 0)$  and  $\Pr(Y \in A, D = 0 \mid Z = 0) \geq \Pr(Y \in A, D = 0 \mid Z = 1)$ . Kitagawa (2015) tests them; rejection means exclusion, independence, or monotonicity failed.
- Passing does not establish monotonicity; for lotteries and distance the case is institutional.

## Can You Check Monotonicity? (2) Bounding the Damage

With defiers present and first stage  $\Delta_D = \Pr(C) - \Pr(D)$ :

$$\text{Wald} = \frac{\Pr(C) \tau_C - \Pr(D) \tau_D}{\Delta_D} \quad \text{numerator and denominator by type}$$

$$\tau_C = \frac{\text{Wald} \cdot \Delta_D + \Pr(D) \tau_D}{\Delta_D + \Pr(D)} \quad \text{solve for the complier effect}$$

Only  $\Pr(D)$  and  $\tau_D$  are unknown. Assume **at most a share  $\delta$  of defiers** and a bounded outcome,  $|\tau_D| \leq M$ ; the worst cases bound  $\tau_C$ .

- **The judge toy example:** Wald = -0.20,  $\Delta_D = 0.10$ ,  $M = 1$ ,  $\delta = 0.20$  gives  $\tau_C \in [-0.73, 0.60]$ : a first stage no larger than the defier share is fatal.
- **Sharp version** (Noack 2021): sensitivity parameters for the defier share and the gap  $\tau_D - \tau_C$ ; report how large they must be before the conclusion flips.
- **Rescue without a bound** (de Chaisemartin 2017): if a subgroup of compliers matches the defiers in size and mean effect, Wald is the LATE of the remaining compliers. Untestable.

# Strength

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## Weak Instruments Are Worse Than No Instrument ★

Recall  $\hat{\beta}_{IV} = \text{Cov}_n(Z, Y) / \text{Cov}_n(Z, D)$ . If the denominator is near zero, small violations of exclusion get *divided by* a small number:

$$\text{plim } \hat{\beta}_{IV} = \beta + \frac{\text{Cov}(Z, \epsilon)}{\text{Cov}(Z, D)}.$$

- An instrument that is one-tenth as endogenous as  $D$  but one-hundredth as strongly correlated with it produces **more** bias than plain OLS.
- The IV standard error is roughly the OLS standard error divided by  $|\rho_{ZD}|$ , the correlation between instrument and treatment:  $\rho_{ZD} = 0.14$  costs a factor of 7.
- Worse, under weak identification  $\hat{\beta}_{IV}$  has **no Gaussian limit**: heavy tails, so the  $t$ -test fails no matter how large  $n$  is.
- The problem *worsens* with more instruments, because many weak instruments overfit the first stage (Bound–Jaeger–Baker, in the Angrist–Krueger frames).

## Diagnosing Instrument Strength ★

- **First-stage  $F$**  on the excluded instruments. The traditional rule of thumb is  $F > 10$  (Staiger and Stock 1997; Stock and Yogo 2005), derived so that IV bias is under 10% of OLS bias.
- **That rule is about bias, not about test size, and it is now known to be far too lax for inference.** Lee, McCrary, Moreira, and Porter (2022) show that a conventional 5%  $t$ -test has correct size only when the first-stage  $F$  exceeds roughly **104**. Their  $\mathfrak{t}F$  procedure replaces the 1.96 critical value with a larger one read off  $F$ : at  $F = 10$  the adjusted critical value is about **3.43**, so the honest interval is roughly **1.75** times wider than the printed one.
- Use a *robust* first-stage  $F$  if you use robust or clustered standard errors in the second stage.
- With several endogenous regressors, use the *Kleibergen–Paap* rank statistic; a strong  $F$  on each instrument separately is not enough.

# Applications

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## Four Canonical Applications

**Goal:** read four classic instruments in one fixed order: the question, the data, the instrument, the three conditions, the main results.

|             | Angrist & Krueger<br>(1991)                 | Card (1995)                         | Angrist (1990)                                                    | AJR (2001)                                |
|-------------|---------------------------------------------|-------------------------------------|-------------------------------------------------------------------|-------------------------------------------|
| Unit        | 329,509 men born<br>1930–39, 1980<br>Census | 3,010 men, NLS<br>Young Men, 1976   | white men born<br>1950–52, Social<br>Security earnings<br>records | 64 former colonies,<br>one cross-section  |
| $D$         | years of schooling                          | years of schooling                  | Vietnam-era<br>military service                                   | protection against<br>expropriation, 0–10 |
| $Y$         | log weekly wage,<br>1979                    | log hourly wage,<br>1976            | FICA earnings, 1981                                               | log GDP per<br>capita, 1995               |
| $Z$         | born in the first<br>quarter                | grew up near a<br>four-year college | draft-eligible lottery<br>number                                  | log settler mortality                     |
| First stage | −0.109 (0.013)                              | 0.337 (0.081)                       | 0.159 (0.040)                                                     | −0.61 (0.13)                              |

## In the Literature: Angrist and Krueger (1991)

**The question.** What does a year of schooling ( $D$ ) do to earnings ( $Y$ , log weekly wage)? Ability raises both, so OLS is biased up; reported schooling is noisy, which pulls OLS down. The two biases have unknown relative size.

**The data.** 1980 Census, 5% sample: 329,509 men born 1930–39 with positive earnings in 1979.

**The instrument.**  $Z$  = born in the first quarter of the year. Children start school in the calendar year they turn six but may leave at a fixed *age*, 16 or 17. A child born in January is older than a classmate born in December in every grade and reaches the leaving age with less schooling completed.

## Angrist and Krueger: the Three Conditions

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|              |                                                                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevance    | First-quarter men have 0.109 fewer years of schooling (0.013). <b>Testable, passes</b> , but the first stage is a tenth of a year.                                                               |
| Independence | Birth quarter is unrelated to earnings potential. Threat: season of birth varies with mothers' income and education and with health at birth, which also move earnings.                          |
| Exclusion    | Birth quarter affects earnings only through years of schooling. Threat: the January child is the oldest in the class; relative age may shape test scores and confidence, and earnings, directly. |

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- The two threats are different: the first puts an arrow *into*  $Z$ , the second an arrow *out of*  $Z$  that bypasses  $D$ .
- **Whose effect.** The men moved by the instrument are those who would have left at the minimum age. The estimate is a return to schooling at the bottom of the distribution, the quantity compulsory attendance laws care about.

*Compliers* in the sense of the previous section: the people whose schooling the instrument actually moved.

## Angrist and Krueger: Main Results

Their Table III, men born 1930–39, first quarter against the other three:

|                           | born Q1 | born Q2–Q4 | difference (se)  |
|---------------------------|---------|------------|------------------|
| log weekly wage           | 5.892   | 5.903      | −0.0111 (0.0027) |
| years of schooling        | 12.688  | 12.797     | −0.109 (0.013)   |
| Wald = $-0.0111 / -0.109$ |         |            | 0.102 (0.024)    |
| OLS                       |         |            | 0.0709 (0.0003)  |

- **Wald is above OLS**, not below: no sign of ability bias. Measurement error in schooling, or a high return for the compliers, can explain the ordering.
- **The price of a small first stage.** The Wald standard error is 80 times the OLS one, from the same 329,509 men. Dividing a tenth of a year into a one-percent wage gap magnifies every error in both.

## Angrist and Krueger: the Cautionary Half

The paper is also the standard illustration of many weak instruments. It combines instruments, which is **2SLS**, the section after the applications, and shows the weak-identification problem at scale.

- To gain precision, the authors interacted quarter of birth with state and year of birth: up to 180 instruments.
- Bound, Jaeger, and Baker (1995): with that many weak instruments, 2SLS is biased *toward OLS*. Replacing quarter of birth with *randomly generated* instruments gave similar estimates and standard errors.
- The mechanism is overfitting the first stage:  $\hat{D}$  reproduces the endogenous variation we were trying to discard.
- **More instruments is not more information.**

Other instruments worth knowing: sibling sex composition for fertility, lottery admission for school quality.

## In the Literature: Card (1995)

**The question.** The return to schooling once more:  $D$  = years of schooling,  $Y$  = log hourly wage in 1976. Same two biases as Angrist and Krueger, ability up and measurement error down.

**The data.** NLS Young Men: 3,010 men first interviewed in 1966, wages in 1976. The data of Lab 7 (`figs/w7_card.R`).

**The instrument.**  $Z$  = grew up in a county with a four-year college (68% did). A nearby college lowers the cost of attending: a student can live at home. The push is strongest for families who could not pay for room and board.

## Card: the Three Conditions

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|              |                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevance    | Growing up near a college adds 0.337 years of schooling (0.081), robust $F = 17.5$ . <b>Testable, passes, not strong.</b>                                                                                        |
| Independence | Colleges sit where schools and labor markets are better, and the families who live there differ. Card conditions on family background; Lab 7 on region and metropolitan status. Argued, conditional on controls. |
| Exclusion    | Proximity affects wages only through schooling. Threat: the same local labor market raises wages directly. The same controls carry this argument.                                                                |

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- Here independence and exclusion fail through the same variable, local economic conditions, so one set of controls serves both.
- **Whose effect.** Men from poorer families for whom a nearby college made attendance possible.

## Card: Main Results

Lab 7's data, controls for experience, experience<sup>2</sup>, Black, metropolitan, South:

|          | OLS     | first stage | reduced form | IV = 0.0446/0.3373 |
|----------|---------|-------------|--------------|--------------------|
| estimate | 0.074   | 0.337       | 0.045        | 0.132              |
| (HC1 se) | (0.004) | (0.081)     | (0.016)      | (0.049)            |

- **IV is nearly twice OLS**, the opposite of ability bias. Card's reading: the compliers have a high return, and measurement error pulls OLS down.
- **Strength.**  $F = 17.5$ ,  $t = 2.72$ : inside the range where the  $t$ -test misleads. Lab 7 computes the Anderson–Rubin set instead.

## In the Literature: Angrist (1990)

**The question.** What did Vietnam-era military service ( $D$ ) do to later earnings ( $Y$ )? Veterans differ from non-veterans in health, schooling, and family background, and enlistment is a choice: OLS compares the wrong men.

**The data.** Social Security (FICA) earnings records for white men born 1950–52, matched to their birthdates' lottery numbers; about 16,000 men per cohort. Veteran status by eligibility from the 1984 SIPP.

**The instrument.**  $Z_i = 1$  if the man's birthdate drew a number below the year's ceiling. The 1970–72 lotteries gave every birthdate a random number; low numbers were called first, so eligible men served more often.

## Angrist: the Three Conditions

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|              |                                                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevance    | Veteran share 0.353 if eligible, 0.193 if not: first stage 0.159 (0.040).<br><b>Testable, passes.</b>                                                      |
| Independence | The number is a lottery draw. <b>Holds by construction:</b> nothing that moves earnings also moves the draw.                                               |
| Exclusion    | Eligibility affects earnings only through service. Threat: eligible men stayed in school to keep deferments, a path from $Z$ to $Y$ that bypasses service. |

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- The cleanest possible independence and still a live exclusion worry: the two conditions are distinct, as the DAG frame said.
- **Whose effect.** Men who served because they were drafted and would not have volunteered.

## Angrist: Main Results

White men born in 1950, 1981 earnings in current dollars:

|          | first stage | reduced form (1981 \$) | Wald                    |
|----------|-------------|------------------------|-------------------------|
| estimate | 0.159       | -435.8                 | $-435.8/0.159 = -2,741$ |
| (se)     | (0.040)     | (210.5)                | about 15% of earnings   |

- **Reading.** Eligibility lowered earnings by \$436 on average across all eligible men; scaled by the 16 points of service it induced, service cost the compliers about \$2,700 a year.
- **Precision.** With a first stage of 0.16 the Wald standard error is about \$1,100 even with 16,000 men per cohort. A clean instrument buys credibility, not precision.

## In the Literature: Acemoglu, Johnson, and Robinson (2001)

**The question.** Do institutions cause national development, measured as GDP per capita?  $D$  = average protection against expropriation risk 1985–95, a 0–10 rating;  $Y$  = log GDP per capita in 1995. OLS fails three ways: rich countries can afford good institutions (simultaneity); culture, geography, and history drive both (omitted variables); the rating is noisy (measurement error, pulling OLS toward zero).

**The data.** One cross-section of 64 former European colonies.

**The instrument.**  $Z$  = log mortality faced by European soldiers, bishops, and sailors in the colonies, 17th–19th centuries. Where settlers could survive, they settled and built inclusive institutions: property rights, rule of law. Where they died, colonization was extractive. Those institutional forms persisted after the disease environment stopped mattering.

## AJR: the Three Conditions

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|              |                                                                                                                                                                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevance    | Higher settler mortality, worse institutions today: first stage $-0.61$ (0.13).<br><b>Testable, passes.</b>                                                                                                                                         |
| Independence | Mortality was set by the diseases Europeans met, not by the colony's later development. Argued from history.                                                                                                                                        |
| Exclusion    | Mortality affects GDP per capita today <i>only</i> through institutions. <b>The leading threat is geography:</b> the diseases that killed settlers also shape agriculture, today's disease burden, and human capital, which affect output directly. |

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- AJR's answer to the threat is an *assumption*: geography is captured by latitude and continent dummies. A functional-form assumption doing identification work, rarely defended (Week 13).
- Later critiques questioned the mortality data itself. **The pattern:** exclusion cannot be tested, so the debate becomes one about history, measurement, and functional form. That is what an IV paper should invite.

## AJR: Main Results

Their Table 4, 64 countries; coefficient on the 0–10 institutions rating:

|                        | OLS    | first stage | reduced form | 2SLS        |
|------------------------|--------|-------------|--------------|-------------|
| base sample            | 0.52   | −0.61       | −0.57        | 0.94        |
| (se)                   | (0.06) | (0.13)      | implied      | (0.16)      |
| with latitude          | 0.47   | −0.51       |              | 1.00 (0.22) |
| with continent dummies | 0.42   | −0.43       |              | 0.98 (0.30) |

- **IV is nearly twice OLS.** AJR read the gap as measurement error in the rating attenuating OLS. One rating point multiplies GDP per capita by  $e^{0.94} = 2.6$ .
- **The geography controls** that the exclusion argument leans on leave the estimate where it was; the first stage weakens and the standard error grows.

## Applications in Sociology

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# Applications in Sociology

**Goal:** the same reading order on sociological questions.

|                 | Sharkey,<br>Torrats-Espinosa &<br>Takyar (2017)           | de Vaan & Stuart (2019)                                      | Aksoy & Gambetta (2022)                                    |
|-----------------|-----------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------|
| Unit            | city $\times$ year; 264 cities,<br>1990–2013              | family $\times$ ED visit;<br>253,407 visits, 2010–15         | province $\times$ election; 81<br>provinces, 1973–2018     |
| $D$             | community nonprofits<br>per 100,000                       | source patient fills an<br>opioid prescription               | fasting hours on the<br>median Ramadan day;<br>religiosity |
| $Y$             | log murder, violent,<br>property crime                    | another household<br>member fills one within a<br>year       | Islamist vote share;<br>Quran-course attendance            |
| $Z$             | arts, medical,<br>environmental nonprofits<br>per 100,000 | assigned ED physician's<br>leave-one-out prescribing<br>rate | daylength = latitude $\times$<br>lunar calendar            |
| First-stage $F$ | 112                                                       | about 820 ( $t = 28.6$ )                                     | daylength is the treatment                                 |

## In the Literature: Sharkey, Torrats-Espinosa, and Takyar (2017)

**The question.** Did community nonprofits cut crime, 1990–2013?  $D$  = nonprofits per 100,000 residents focused on crime, violence, and community development;  $Y$  = log murder, violent, and property crime. OLS fails: nonprofits form where crime rises (simultaneity, pushing OLS toward zero) and where civic capacity is high (omitted variable); the IRS registry counts organizations with error.

**The data.** 264 cities with 100,000 or more residents, 1990–2013. Nonprofits from the IRS registry via the National Center for Charitable Statistics, crime from the Uniform Crime Reports. Long differences, and city and year fixed effects.

**The instrument.**  $Z$  = nonprofits per 100,000 for the arts, medical research, and the environment: organizations that share the city's funding climate and do not target violence.

# Sharkey, Torratts-Espinosa, and Takyar: Conditions

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|              |                                                                                                                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevance    | Cities where arts and medical nonprofits grow also grow community nonprofits: first stage 0.607 (0.057), $F = 112$ in long differences; 0.584 (0.049), $F = 143$ with fixed effects. <b>Testable, strong.</b> |
| Independence | A city's philanthropic climate is unrelated to its crime trend, given the controls. Threat: a philanthropic boom may signal the prosperity that cuts crime; the city-level controls carry the argument.       |
| Exclusion    | Arts, medical, and environmental nonprofits affect crime only through community nonprofits. Threat: an arts program may occupy youth or improve a block on its own.                                           |

---

- A strong first stage settles relevance and nothing else; the other two conditions rest entirely on argument.

## Sharkey, Torratts-Espinosa, and Takyar: Results

Their Tables 2–3, effect of ten more community nonprofits per 100,000 residents:

|     | long differences |         |          | city and year fixed effects |         |          |
|-----|------------------|---------|----------|-----------------------------|---------|----------|
|     | murder           | violent | property | murder                      | violent | property |
| OLS | −6%              | −6%     | −5%      | −5%                         | −4%     | −4%      |
| IV  | −12%             | −10%    | −7%      | −9%                         | −6%     | −4%      |

- **IV exceeds OLS** in every column: measurement error in the registry attenuates OLS, or this is the effect in cities whose nonprofit sector responds to the funding climate.
- **Ordering.** Largest for murder, smallest for property crime, under both estimators and both designs.

## In the Literature: de Vaan and Stuart (2019)

**The question.** Does opioid use spread inside the family?  $D$  = a family member, the *source patient*, fills an opioid prescription after an emergency department (ED) visit;  $Y$  = another member of the household fills one within a year. OLS fails: households share genes, partners sort on health behavior, and members share conditions and physicians.

**The data.** Massachusetts All Payer Claims Database, 2010–2015, every insured resident's medical and pharmacy claims; families read off shared insurance policies. Sample: 253,407 ED visits by a member of a family with no prior opioid use; 13% of the visits ended in an opioid prescription.

**The instrument.**  $Z$  = the opioid prescribing rate of the ED physician the source patient happened to see, computed on that physician's other visits (a leave-one-out mean over 1,075 physicians). Arriving patients get whoever is on duty.

## de Vaan and Stuart: the Three Conditions

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|              |                                                                                                                                                                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevance    | One SD more prescribing raises the source patient's chance of a prescription from 13.0% to 15.3%: first stage 0.859 (0.030) with ED and month fixed effects and controls. <b>Testable, very strong.</b>                                                                    |
| Independence | Within an ED and month, the on-duty physician is as good as randomly assigned. Balance (their Table 4): patients of high and low prescribers match on prior benzodiazepine use (.38 / .38), comorbidity index (1.08 / 1.08), sex (.50 / .50); age differs by three months. |
| Exclusion    | The physician's habit reaches the household only through the source patient's prescription. Threat: a liberal prescriber may differ in other advice or drugs. Defense: neither patient nor family sees an ED physician again, so the visit is the only contact.            |

---

- This is an **examiner design**: the leave-one-out leniency of a randomly assigned decision-maker. A later section builds it from scratch and adds the condition it needs beyond these three.

## de Vaan and Stuart: Main Results

Their Tables 5–6, family level, outcome within a year; base rate 14.6%:

|                  | OLS, raw | OLS, FE and controls | first stage | IV      |
|------------------|----------|----------------------|-------------|---------|
| estimate         | 0.023    | 0.027                | 0.859       | 0.130   |
| (se)             | (0.002)  | (0.002)              | (0.030)     | (0.038) |
| relative to base | +16%     | +18%                 |             | +95%    |

- **IV is five times OLS.** The authors' reading: the instrument moves only households that fill what is prescribed, and the effect is large there; OLS averages in families who refuse opioids whatever the physician does. Prescriptions written but never filled are unobserved, which also pulls OLS down.
- **Robustness.** 24 OLS and 24 IV specifications (three windows, two levels, four control sets): IV effects run from +75% to +200%, every one significant.

## In the Literature: Aksoy and Gambetta (2022)

**The question.** Does a costlier religious practice strengthen religiosity and political Islam? The cost is the length of the Ramadan fast.  $D$  = fasting hours on the median Ramadan day;  $Y$  = Islamist party vote share, and Quran-course attendance. OLS fails: provinces that fast devoutly are also poorer and more rural (omitted variables).

**The data.** 81 Turkish provinces  $\times$  elections, 1973–2018, with province and year fixed effects; a 2014 individual survey for the mechanism step.

**The instrument.** Daylength on the median Ramadan day = latitude  $\times$  the lunar calendar. Ramadan moves back 11 days a year, so summer Ramadans mean long fasts, more so at high latitude. The fixed effects leave only the interaction.

## Aksoy and Gambetta: the Three Conditions

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|              |                                                                                                                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevance    | Daylength <i>is</i> the length of the fast. Holds by construction.                                                                                                                                                         |
| Independence | Given province and year effects, the latitude $\times$ lunar calendar interaction is unrelated to anything else provincial. Placebo outcomes, turnout and population: null.                                                |
| Exclusion    | Needed only for the mechanism step, daylength $\rightarrow$ religiosity $\rightarrow$ vote. Threat: a longer fast also lowers growth (Campante and Yanagizawa-Drott 2015), a path from $Z$ to $Y$ not through religiosity. |

---

When the instrument is the variable of interest, the reduced form is the causal estimate, and exclusion is not required for it. The IV step asks a second question, about mechanism.

## Aksoy and Gambetta: Main Results

The main model is the reduced form,  $Y_{it} = \mu_i + \delta_t + \alpha R_{it} + \epsilon_{it}$ ,  $R_{it}$  the fasting hours:

| per hour of fasting                                                | estimate | (se)    |
|--------------------------------------------------------------------|----------|---------|
| Islamist vote share, points                                        | 7.16     | (2.54)  |
| Quran-course attendance, per 1,000 residents                       | 1.76     | (0.76)  |
| SEM total effect on AKP vote, per daylength category (2014 survey) | 0.03     | (0.009) |

- **Reading.** A half-hour longer fast adds 3.5 points of vote share, 11% of the 31% mean.
- **The IV step** instruments latent religiosity with daylength in the survey; it says the vote moves *because* religiosity moves, if exclusion holds.

## 2SLS

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## The Two Stages ★

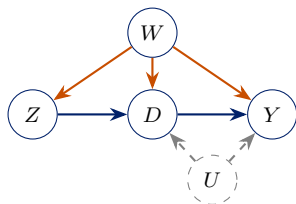
With covariates and possibly several instruments:

$$\text{Stage 1: } D_i = \pi_0 + \pi_1 Z_i + \pi_2' W_i + v_i \implies \hat{D}_i$$

$$\text{Stage 2: } Y_i = \alpha + \beta \hat{D}_i + \gamma' W_i + u_i$$

- $\hat{D}_i$  is the part of  $D_i$  *explained by the instrument*, which by assumption is uncorrelated with  $\epsilon_i$ . We regress on the clean part and discard the rest.
- **The exogenous covariates  $W_i$  must appear in both stages.** Omitting them from the first stage is a common and serious error.
- **One excluded instrument per endogenous regressor (*just identified*)**, the case this course uses throughout.
- **Never run the two stages by hand.** The second-stage standard errors would treat  $\hat{D}$  as data rather than as an estimate. Use `ivreg()` or `feols(y ~ w | d ~ z)`.

## Why $W$ Goes in Both Stages



$Z \leftarrow W \rightarrow Y$  is a back door from instrument to outcome:  $Z$  is valid only *given*  $W$ . 2SLS with  $W$  in both stages partials  $W$  out of  $Y$ ,  $D$ ,  $Z$ , then takes the one- $Z$  Wald ratio (FWL).  $W$  belongs in the first stage when it affects  $Z$ , in the second when it affects  $Y$ .

**Simulation**, true  $\beta = 1$ ,  $n = 5,000$ , 500 draws (figs/w7\_twostage\_W.R); mean  $\hat{\beta}$ :

| $W$ enters        | $W \rightarrow Z$ | $Z \perp W$ |
|-------------------|-------------------|-------------|
| both stages       | 1.00              | 1.00        |
| second stage only | 0.56              | 1.00        |
| first stage only  | 1.71              | 1.90        |
| neither           | 1.55              | 1.00        |

- **Second stage only.**  $\hat{D}$  from  $Z$  alone carries  $W$ 's variation, and  $D - \hat{D}$  is correlated with  $W$  in the second-stage error. Harmless only if  $Z \perp W$  (a lottery).
- **First stage only.**  $\hat{D}$  contains  $W$ , and  $W \rightarrow Y$  is in the error: exclusion fails for  $\hat{D}$ .

## IV Standard Error, Scalar Version (1)

One  $Y$ , one  $D$ , one  $Z$ . Substitute the model for  $Y_i$  into the Wald ratio:

$$\begin{aligned}\hat{\beta}_{IV} &= \frac{\text{Cov}_n(Z_i, Y_i)}{\text{Cov}_n(Z_i, D_i)} = \frac{\text{Cov}_n(Z_i, \alpha + \beta D_i + \epsilon_i)}{\text{Cov}_n(Z_i, D_i)} && \text{substitute } Y_i = \alpha + \beta D_i + \epsilon_i \\ &= \beta + \frac{\text{Cov}_n(Z_i, \epsilon_i)}{\text{Cov}_n(Z_i, D_i)} && \text{covariance is linear; the constant drops} \\ &= \beta + \frac{\sum_i \tilde{z}_i \epsilon_i}{\sum_i \tilde{z}_i \tilde{d}_i} && \text{centered } Z \text{ and } D; \text{ the } 1/n \text{ cancel}\end{aligned}$$

- **The numerator is the random part:** independent terms  $\tilde{z}_i \epsilon_i$  with mean zero and variance  $\tilde{z}_i^2 \sigma_i^2$ ; its variance is their sum.
- **The denominator is nearly fixed:**  $\frac{1}{n} \sum_i \tilde{z}_i \tilde{d}_i$  settles at  $\text{Cov}(Z, D)$  as  $n$  grows (law of large numbers). If  $\text{Cov}(Z, D)$  is near zero, its small changes matter: the weak-IV problem.

$$\text{Var}(\hat{\beta}_{IV}) \approx \frac{\sum_i \tilde{z}_i^2 \sigma_i^2}{(\sum_i \tilde{z}_i \tilde{d}_i)^2}.$$

## IV Standard Error, Scalar Version (2)

Take  $\sigma_i^2 = \sigma^2$  for every  $i$  and divide top and bottom by  $\sum_i \tilde{z}_i^2 \sum_i \tilde{d}_i^2$ ;  $\rho_{ZD}$  is the correlation of  $Z$  and  $D$ :

$$\text{Var}(\hat{\beta}_{IV}) = \frac{\sigma^2 \sum_i \tilde{z}_i^2}{(\sum_i \tilde{z}_i \tilde{d}_i)^2} = \frac{\sigma^2}{\sum_i \tilde{d}_i^2} \cdot \frac{\sum_i \tilde{z}_i^2 \sum_i \tilde{d}_i^2}{(\sum_i \tilde{z}_i \tilde{d}_i)^2} = \underbrace{\frac{\sigma^2}{n s_D^2}}_{\text{OLS variance}} \cdot \frac{1}{\rho_{ZD}^2}$$

- **IV standard error = OLS standard error /  $|\rho_{ZD}|$** , for the same  $\sigma$ . The instrument explains a fraction  $\rho_{ZD}^2$  of the variance in  $D$ , and only that fraction is used:  $\rho_{ZD} = 0.14$  costs a factor of 7.
- **Which residual.**  $\sigma^2$  is estimated from  $\hat{\epsilon}_i = Y_i - \hat{\alpha} - \hat{\beta}_{IV} D_i$ , with the *actual*  $D_i$ . A second stage run by hand uses  $\hat{D}_i$  instead, gets a smaller residual, and prints too small a standard error.
- **Robust version.** Keep  $\sigma_i^2$  and plug in  $\hat{\epsilon}_i^2$ :  $\widehat{\text{Var}} = \sum_i \tilde{z}_i^2 \hat{\epsilon}_i^2 / (\sum_i \tilde{z}_i \tilde{d}_i)^2$ .

## The Scalar Formula on Card's Data

Log wage on schooling, instrument near a four-year college, no controls ( $n = 3,010$ )  
(figs/w7\_card.R).

|                           | estimate | standard error                           |
|---------------------------|----------|------------------------------------------|
| OLS                       | 0.052    | 0.00287                                  |
| IV, homoskedastic formula | 0.188    | $0.00287 \times 1.321 / 0.1442 = 0.0263$ |
| IV, robust formula        | 0.188    | 0.0261 (ivreg + vcovHC: 0.0261)          |
| second stage by hand      | 0.188    | 0.0207 too small                         |

- $\rho_{ZD} = 0.144$ : proximity explains 2% of the variance in schooling, so  $1/|\rho_{ZD}| = 6.9$ .
- The IV residual is larger than the OLS residual (sd 0.557 against 0.421) because the IV slope is not the least-squares slope: another factor of 1.32. Together, 9.2 times the OLS standard error.
- By hand, the second-stage residual has sd 0.438, not 0.557: 21% too small.

## IV in Matrix Form ★

Start from the moment condition  $E[\mathbf{Z}'u] = \mathbf{0}$  with  $Y = D\beta_0 + u$ . The sample analogue  $\mathbf{Z}'(\mathbf{Y} - \mathbf{D}\hat{\beta}) = \mathbf{0}$  has no solution when there are more instruments than regressors, so instead *minimize* the sample moments in a weighted norm:

$$\hat{\beta} = \arg \min_b (\mathbf{Z}'(\mathbf{Y} - \mathbf{D}b))'(\mathbf{Z}'\mathbf{Z})^{-1}(\mathbf{Z}'(\mathbf{Y} - \mathbf{D}b)).$$

The first-order condition gives

$$\hat{\beta} = \left( \mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D} \right)^{-1} \mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{Y}.$$

## IV in Matrix Form: The 2SLS Estimator ★

Define the projection matrix  $\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$ , the same object as Week 1, now projecting onto the instrument space:

$$\hat{\beta}_{2SLS} = (\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1}\mathbf{D}'\mathbf{P}_Z\mathbf{Y}$$

Since  $\mathbf{P}_Z\mathbf{D}$  is the first-stage fitted value and  $\mathbf{P}_Z$  is idempotent, this is literally “regress  $\mathbf{Y}$  on  $\hat{\mathbf{D}}$ .” And when  $\mathbf{Z} = \mathbf{D}$ ,  $\mathbf{P}_Z\mathbf{D} = \mathbf{D}$  and 2SLS collapses to OLS: OLS is the special case in which you claim everything is exogenous.

## The IV Sandwich ★

Substitute  $\mathbf{Y} = \mathbf{D}\beta_0 + \mathbf{u}$ :

$$\hat{\beta} - \beta_0 = (\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1}\mathbf{D}'\mathbf{P}_Z\mathbf{u}.$$

Multiply by  $\sqrt{n}$ . Under regularity conditions and a *strong* instrument,

$$\frac{1}{n}\mathbf{D}'\mathbf{P}_Z\mathbf{D} \xrightarrow{p} \mathbf{A} \succ 0, \quad \frac{1}{\sqrt{n}}\mathbf{D}'\mathbf{P}_Z\mathbf{u} \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{S}),$$

so by Slutsky  $\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{A}^{-1}\mathbf{S}\mathbf{A}^{-1})$ .

The robust variance estimator is the familiar sandwich:

$$\hat{V}(\hat{\beta}) = (\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1}\mathbf{D}'\mathbf{P}_Z\hat{V}(\mathbf{u})\mathbf{P}_Z\mathbf{D}(\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1},$$

plugging in  $\hat{V}(\mathbf{u}) = \text{diag}(\hat{u}_i^2)$ , and clustering exactly as in Week 2. **Note the residuals  $\hat{u}_i$  must be computed with the actual  $D_i$ , not  $\hat{D}_i$** ; this is the other reason not to run the stages by hand.

## Weak IV

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# The Goal

**Goal:** a weak first stage breaks the  $t$ -test: the usual 95% CI,  $\hat{\beta}_{IV} \pm 1.96 \widehat{SE}$ , though wide, covers the truth less than 95% of the time, and  $\hat{\beta}_{IV}$  has no Gaussian limit.

**Anderson–Rubin** tests a guessed  $\beta_0$  without ever dividing by the first stage: it repairs *inference*; the estimate is unchanged.

## Plan

1. The equation to test.
2. Two ways to use it: solve it or check it.
3. The statistic.
4. The confidence region, on Card's data and in simulation.

## Anderson–Rubin, Step 1: the Equation to Test

The IV assumption: the instrument is unrelated to the error,  $\text{Cov}(Z_i, \epsilon_i) = 0$ .  
Substitute what the error is:

$$\epsilon_i = Y_i - \alpha - \beta D_i \quad \text{the structural equation, solved for } \epsilon_i$$

$$0 = \text{Cov}(Z_i, \epsilon_i) = \text{Cov}(Z_i, Y_i - \beta D_i) \quad \alpha \text{ is a constant}$$

$$= \text{Cov}(Z_i, Y_i) - \beta \text{Cov}(Z_i, D_i) \quad \text{covariance is linear}$$

- **In words:** strip the treatment's part,  $\beta D$ , out of  $Y$ ; what remains must be unrelated to  $Z$ . The true  $\beta$  satisfies this equation.
- **Solve it for  $\beta$**  and you get the Wald estimator,  $\text{Cov}(Z, Y) / \text{Cov}(Z, D)$ . Nothing new is assumed.

## Anderson–Rubin, Step 2: Solve It or Check It

**Solve it** (Wald, 2SLS):

$\hat{\beta}_{IV} = \text{Cov}_n(Z, Y) / \text{Cov}_n(Z, D)$ . Divides by the first stage; unreliable when that is small and noisy.

**Check it** (Anderson–Rubin): guess  $\beta_0$ , ask whether  $\text{Cov}_n(Z, Y - \beta_0 D) = 0$ . Never divides.

**Card's data** ( $Z$  = near a college,  $D$  = schooling,  $Y$  = log wage):

$\text{Cov}_n(Z, Y) = 0.0338$ ,  $\text{Cov}_n(Z, D) = 0.1798$  (figs/w7\_weakiv.R).

| guess $\beta_0$                                                   | 0     | 0.10  | 0.144 | 0.188 | 0.25   | 0.30   |
|-------------------------------------------------------------------|-------|-------|-------|-------|--------|--------|
| $\text{Cov}_n(Z, Y - \beta_0 D) = 0.0338 - \beta_0 \times 0.1798$ | 0.034 | 0.016 | 0.008 | 0.000 | -0.011 | -0.020 |

Zero at  $\beta_0 = 0.188$ , the Wald estimate. For the other guesses: how far from zero is too far? That needs a scale.

## Anderson–Rubin, Step 3: the Statistic

1. Guess  $\beta_0$ ; build  $U_i = Y_i - \beta_0 D_i$ .
2. Regress  $U_i$  on  $Z_i$ . Since  $U_i = \alpha + (\beta - \beta_0)D_i + \epsilon_i$ , the true slope is  $(\beta - \beta_0)\pi$ : zero if the guess is right.
3.  $C(\beta_0) = t^2$  of that slope;  $\chi_1^2$  under  $H_0: \beta = \beta_0$ . Reject the guess if  $C(\beta_0) > 3.84$ .

| guess $\beta_0$     | 0      | 0.10   | 0.144 | 0.188 | 0.25   | 0.30   |
|---------------------|--------|--------|-------|-------|--------|--------|
| slope of $U$ on $Z$ | 0.156  | 0.073  | 0.037 | 0.000 | -0.051 | -0.093 |
| $t$                 | 9.10   | 4.25   | 1.91  | 0.00  | -1.94  | -3.02  |
| $C(\beta_0) = t^2$  | 82.7   | 18.0   | 3.66  | 0.00  | 3.76   | 9.12   |
| verdict             | reject | reject | keep  | keep  | keep   | reject |

Step 2 is an ordinary regression  $t$ -test:  $\hat{\pi}$  never appears in a denominator. A weak instrument shrinks  $(\beta - \beta_0)\pi$  for every guess, so fewer guesses are rejected: a wider set, not a wrong one.

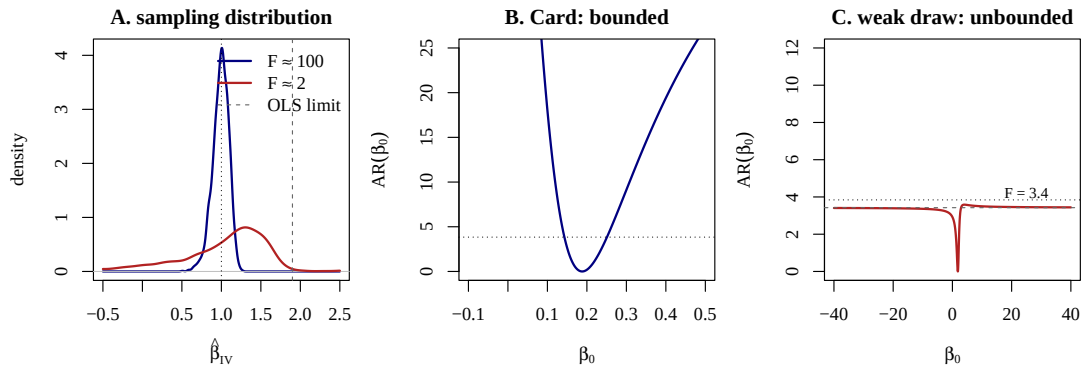
## Anderson–Rubin, Step 4: the Confidence Region

Keep every guess the data cannot reject,  $CR(\beta) = \{\beta_0 : C(\beta_0) \leq c_{1-\alpha}\}$ . “**Works under a weak instrument**” means: covers the truth 95% of the time whatever  $F$  is. Simulation,  $n = 500$ ,  $\beta = 1$ , 4,000 draws (figs/w7\_weakiv.R):

| median first-stage $F$          | 1.8  | 6.0  | 9.7  | 101  |
|---------------------------------|------|------|------|------|
| usual 95% CI covers the truth   | 87%  | 91%  | 92%  | 95%  |
| AR region covers the truth      | 94%  | 95%  | 95%  | 95%  |
| AR region is a bounded interval | 26%  | 69%  | 87%  | 100% |
| median width, usual CI          | 2.76 | 1.59 | 1.26 | 0.39 |
| median width, AR when bounded   | 2.29 | 2.15 | 1.70 | 0.40 |

- **Why the usual CI fails:** it widens as the instrument weakens, but not enough (87%); it misses the draws that lean toward OLS and look most precise.
- **What AR does:** at  $F \approx 6$ , bounded in 69% of draws, unbounded otherwise; coverage stays at 95%.

# Anderson–Rubin on Card's Data and on the Weak Draw



A: sampling distribution of  $\hat{\beta}_{IV}$  with a strong and a weak instrument, 4,000 draws each. B and C:  $C(\beta_0)$  against  $\beta_0$ ; the dotted line is 3.84 and the dashed line the first-stage  $F$ . The AR region is where the curve lies below 3.84: an interval for Card, the whole line for the weak draw (figs/w7\_weakiv.R).

## When Is AR Useful?

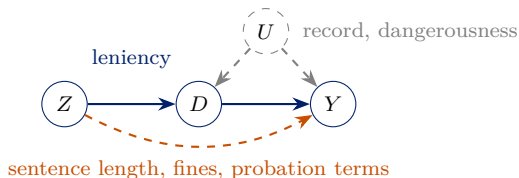
- **Hopeless instrument** ( $F$  below about 4): AR is mostly unbounded. No method can do better; the usual 95% CI prints a finite interval that looks informative and is not. AR refuses.
- **Weak instrument** ( $F$  about 5–20; Card 10, Derenoncourt 15): AR is usually a bounded interval with correct coverage. In the simulation, bounded in 69% of draws at  $F \approx 6$  and 87% at  $F \approx 10$ , coverage 95%; the usual CI covers 91–92%.
- **It can be more informative.** Card's data with controls,  $F \approx 10$  (lab 7): usual CI  $[-0.003, 0.190]$ , includes zero; AR  $[0.004, 0.256]$ , excludes it. AR follows the data's shape instead of a symmetric interval around a biased estimate.

## Judge IV

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## The Design

- **Goal:** among people *convicted of a felony*, the effect of a prison sentence ( $D = 1$ ) rather than probation ( $D = 0$ ) on later reoffending or employment  $Y$ . The pool is the convicted, not the public.
- **Problem:** within that pool, judges send the worse records and graver offenses to prison, and the same things predict  $Y$ ; prisoners and probationers are not comparable.
- **Variation:** cases are assigned to judges at random within a court and year, and judges differ persistently in how often they imprison.  $Z_i$  = the leniency of the judge defendant  $i$  drew.



## Constructing the Instrument

Leave-one-out leniency: your judge's incarceration rate in everyone else's cases (within court and year),  $Z_i = \frac{1}{n_{j(i)}-1} \sum_{k \neq i: j(k)=j(i)} D_k$ .

- **Why leave your own case out.** The leave-in mean contains  $D_i$ , which is correlated with  $\epsilon_i$ ; the instrument inherits  $\text{Cov}(D_i, \epsilon_i)/n_j$  of it, shrinking with cases per judge only.
- **Judge dummies are the leave-in mean in disguise.** With  $J - 1$  judge indicators as instruments (Harding et al.; Kling 2006) the fitted value is the judge's rate over *all* cases, own case included: same bias.

**Simulation** (figs/w7\_judge.R): 100 judges, a confounder raising  $D$  and  $Y$ , true  $\tau = 0.20$ , 500 replications.

| cases per judge | OLS  | judge dummies = leave-in (identical every time) | leave-one-out  |
|-----------------|------|-------------------------------------------------|----------------|
| 50              | 1.17 | 0.56                                            | 0.17 (sd 0.29) |
| 200             | 1.17 | 0.33                                            | 0.20 (sd 0.12) |

## The Three Conditions, Judge Version

- **Relevance.** Judges differ and courts have thousands of cases;  $F$  is rarely the problem. Compute  $Z$  and  $F$  within court  $\times$  year.
- **Independence.** Random assignment is an institutional fact and partly testable: regress  $Z$  on defendant and case characteristics (race, age, priors, offense); nothing should predict the draw. Michigan assigns by a random number generator (Harding et al. 2019).
- **Exclusion.** The harsh judge must change  $Y$  only through prison versus probation. The same judge also sets sentence length, fines, and probation conditions, and speaks differently from the bench: each is a path from  $Z$  to  $Y$ . Remedies: instrument every dimension of the sentence at once (Harding et al.: prison, jail, jail with probation, three sentence lengths), or define the treatment as the whole sentence package.
- **Monotonicity.** With two judges it reads “no defiers.” With many judges it must hold for every pair of judges and every defendant.

# Monotonicity With Many Judges: a Worked Failure

Two judges, two case types in equal shares.

| share incarcerated | drug cases (50%) | property cases (50%) | all cases |
|--------------------|------------------|----------------------|-----------|
| Judge A            | 0.90             | 0.30                 | 0.60      |
| Judge B            | 0.50             | 0.50                 | 0.50      |

A is harsher overall, so  $Z = 1$  for A. Among drug cases A creates 0.40 compliers; among property cases A is the lenient one and creates 0.20 **defiers**.

$$\text{first stage} = 0.5(0.40) - 0.5(0.20) = 0.10 \quad \text{defiers subtract}$$

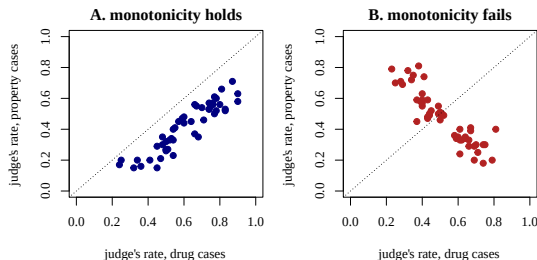
$$\text{reduced form} = 0.5(0.40) \tau_{\text{drug}} - 0.5(0.20) \tau_{\text{prop}} \quad \text{numerator of the LATE derivation}$$

$$\text{Wald} = 2 \tau_{\text{drug}} - \tau_{\text{prop}} \quad \text{weights sum to one; one is negative}$$

With  $\tau_{\text{drug}} = 0.10$  and  $\tau_{\text{prop}} = 0.40$  the Wald estimand is  $-0.20$ : every effect is positive, the IV estimand is negative. Within each case type the ordering is stable and the Wald ratio recovers that type's  $\tau$ .

## Detecting a Monotonicity Failure

If judge A is harsher than B for one kind of defendant, A must be harsher for every kind: plot each judge's rate on one case type against the other (Dobbie, Goldin, and Yang 2018; `figs/w7_mono.R`).



- **Holds:** judges line up alike on both types; correlation of the two rates 0.92, 88% of judge pairs ordered the same way.
- **Fails:** harsh on drugs means lenient on property; correlation  $-0.90$ , 15% of pairs alike. The pooled first stage still looks strong.

## Addressing a Monotonicity Failure

- **The common fix: build leniency within strata.** Compute  $Z_i$  from the judge's other cases of the *same type* as case  $i$  (case type  $\times$  court  $\times$  year), and let the first stage differ by stratum. Within a stratum judges are consistently ordered, the Wald ratio recovers that stratum's  $\tau$ , and 2SLS pools the strata with positive weights.
- **The common assumption: average monotonicity** (Mogstad, Torgovitsky, and Walters 2021). 2SLS with a leniency instrument is a positively weighted average of effects if, for every defendant, leniency and that defendant's own treatment covary non-negatively across judges. Judges may disagree on some cases; no defendant's treatment may run against leniency on average. Plausible when judge differences are mostly a common harshness, as in panel A.
- **A formal test** (Frandsen, Lefgren, and Leslie 2023): with a bounded outcome, exclusion and monotonicity bound the slope of the per-judge mean outcome against the per-judge incarceration rate by the range of  $Y$ . `ivcheck::iv_testjfe` in R.

## Whose Effect?

- **Compliers are the marginal defendants:** those whose sentence depends on which judge they drew. Not the mandatory minimums, not first-degree murder, not the obvious probation cases. In the toy example the pairwise complier share is the first stage, 0.10.
- With a continuous  $Z$ , 2SLS averages pairwise LATEs across the leniency distribution, more weight where judges are dense.
- **Describe them.** Complier means of covariates are recoverable by Abadie's  $\kappa$  reweighting; judge papers report a table of complier characteristics next to the full sample.
- **For once the LATE is the policy parameter.** A sentencing reform moves exactly the marginal cases, not the always-incarcerated.
- **Different margins, different compliers.** Prison versus probation at sentencing (Harding et al. 2017) and detention versus release before trial (Dobbie, Goldin, and Yang 2018) are different populations; their LATEs need not agree.

## In the Literature: Harding, Morenoff, Nguyen, and Bushway

**Data:** everyone sentenced for a felony in Michigan, 2003–2006 ( $n = 111,110$ ), administrative records to 2013–2015; cases with no judicial discretion dropped.

**Instruments:** judge dummies and judge  $\times$  presentence-characteristic interactions in 2SLS, six first stages.

- **PNAS 2017.** Prison rather than probation raises the chance of a return to prison within five years of *release* by about 20 points, for nonwhite and white defendants alike. Most of it is **parole**: returns for technical violations (missed meetings, failed drug tests, curfew), 18 and 14 points; on a new felony sentence, 4 and 9. No significant effect on new felony convictions.
- **AJS 2018.** Prison lowers employment mainly by keeping people *inside*, the sentence itself and the later returns. After release, penalties fall on whites with a presentence work history.

Prison begets prison through parole supervision, not through more crime.

## Shift-share

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## The Problem and the Instrument

**Goal:** the effect of an actual local change  $D_\ell$  (employment growth, immigrant inflow, growth in import exposure) on a local outcome  $Y_\ell$  (wages, marriage, votes), one observation per place  $\ell$ .  $D_\ell$  responds to local conditions, a plant closing or a housing boom, that also move  $Y_\ell$ .

**The instrument:** what  $D_\ell$  would have been if every industry in  $\ell$  had moved at the *national* rate,

$$Z_\ell = \sum_{k=1}^K s_{\ell k} g_k, \quad s_{\ell k} = \text{industry } k\text{'s share of } \ell \text{ at baseline; } g_k = \text{national change in } k.$$

$Z_\ell$  instruments  $D_\ell$ , the local change. The  $g_k$  are not instruments: each is one number for the whole country.  $Z_\ell$  varies across places only because the industry mix  $s_{\ell k}$  does.

**Plan:** what  $Z_\ell$  keeps; three places by hand; two routes to validity; Autor, Dorn, and Hanson (2019), Derenoncourt (2022).

# What $Z_\ell$ Keeps and What It Throws Away

$$\begin{aligned}
 D_\ell &= \sum_k s_{\ell k} g_{\ell k} && \text{actual change} = \text{local industry changes } g_{\ell k}, \text{ weighted} \\
 &= Z_\ell + \underbrace{\sum_k s_{\ell k} (g_{\ell k} - g_k)}_{\text{local part}} && \text{add and subtract the national } g_k
 \end{aligned}$$

The local part is where endogeneity lives (a plant closes because  $\ell$ 's wages fell);  $Z_\ell$  drops it and keeps the national part, which  $\ell$  did not cause.

|                            | $k$      | $D_\ell$ , measured                               | $Z_\ell$ , constructed                                             | $Y_\ell$                           |
|----------------------------|----------|---------------------------------------------------|--------------------------------------------------------------------|------------------------------------|
| Bartik (1991)              | industry | employment growth in $\ell$                       | growth at national industry rates, 1970s shares                    | wage growth                        |
| Autor, Dorn, Hanson (2013) | industry | decade change in US imports from China per worker | same with China's exports to 8 other rich countries, lagged shares | change in manufacturing employment |

## Two Routes to Validity

**Goal:**  $Z_\ell = \sum_k s_{\ell k} g_k$  must be unrelated to everything else that moves  $Y_\ell$ . It is a product of two pieces, so it is enough that *one* of them is as good as random.

**In ADH (2013)**,  $Y_\ell$  = the change in a commuting zone's manufacturing employment, 1990–2007, and

- $s_{\ell k}$  = industry  $k$ 's share of  $\ell$ 's employment ten years before the period (1980 for the 1990s), 397 industries;
- $g_k$  = the growth of China's exports in industry  $k$  to eight other rich countries.

### Plan

1. Route 1, **exogenous shares** (Goldsmith-Pinkham, Sorkin, and Swift 2020): the claim, how it fails, how to check.
2. Route 2, **exogenous shocks** (Borusyak, Hull, and Jaravel 2022): the same three steps.

## Route 1: Exogenous Shares, in ADH

**The claim.** Which places held which industries in 1980 is as good as random, given the controls. Then  $Z_\ell$  is valid whatever the shocks are.

**How it fails, hypothetically.** Furniture and textile towns had older plants and were already losing factory jobs in the 1980s, for reasons unrelated to China. They get a high  $Z_\ell$  and keep losing jobs, and 2SLS credits China with a decline already under way.

**How to check: pre-trends by place.** Regress a *past* change in  $\ell$ 's manufacturing employment on its *future* exposure. ADH, Table 2:

| change in manufacturing employment | coefficient on exposure |
|------------------------------------|-------------------------|
| 1970s, on future exposure          | 0.43 (0.15)             |
| 1980s, on future exposure          | -0.13 (0.13)            |
| 1990–2007, on current exposure     | -0.75 (0.07)            |

Places hit later were not already losing manufacturing.

## Route 2: Exogenous Shocks, in ADH

**The claim.** Across the 397 industries, where China's exports grew is as good as random, driven by China's productivity and WTO entry. Then the shares may be endogenous: furniture towns can differ in every way, as long as the industries China grew in are not the ones in trouble anyway.

**How it fails, hypothetically.**

- China grows most in labor-intensive industries that were already shrinking in every rich country (automation, earlier offshoring).
- Demand rises for a product in all rich countries at once; ADH name consumer electronics. Then  $g_k$  carries US demand after all.

**How to check: pre-trends by industry.** One row per industry: regress the 1980s change in US employment in  $k$  on  $g_k$ . A slope near zero indicates that the industries in which China's exports to the eight other countries grew fastest were not already shrinking in the US.

## Three Places, Two Industries, by Hand

National changes over the decade: manufacturing  $g_M = -10\%$ , services  $g_S = +5\%$ .

Baseline employment shares:

| place | $s_{\ell M}$ | $s_{\ell S}$ | $Z_\ell = s_{\ell M}g_M + s_{\ell S}g_S$ |
|-------|--------------|--------------|------------------------------------------|
| A     | 0.6          | 0.4          | $0.6(-10) + 0.4(5) = -4$                 |
| B     | 0.2          | 0.8          | $0.2(-10) + 0.8(5) = +2$                 |
| C     | 0.4          | 0.6          | $0.4(-10) + 0.6(5) = -1$                 |

- The shocks are common to every place;  $Z_\ell$  varies only through the shares. The design compares A with B: the place that was manufacturing when manufacturing collapsed.
- If A's employment actually fell 7% (a local plant closed too), then  $D_A = -7$ ,  $Z_A = -4$ , and the extra  $-3$  is the local part, discarded.
- 2SLS of  $Y_\ell$  on  $D_\ell$  with  $Z_\ell$  as the one excluded instrument, controls in both stages.  $K$  industries are not  $K$  instruments; what  $K$  changes is the inference.

## In the Literature: Autor, Dorn, and Hanson (2019)

The title is borrowed from Wilson (1996): does the collapse of manufacturing lower young men's value in the marriage market?

- **The change in exposure:**  $D_\ell = \sum_k s_{\ell k} \Delta IP_k$ , with  $s_{\ell k}$  = industry  $k$ 's share of  $\ell$ 's employment in 1990 and  $\Delta IP_k$  = the decade change in US imports from China in  $k$ , divided by US absorption of  $k$  in 1991 (shipments plus net imports).
- **Instrument:** the same sum with the change in China's exports to eight other high-income countries, 1980 employment shares, and absorption lagged three years. 722 commuting zones, 1990–2000 and 2000–2014 changes stacked, SEs clustered by state.
- **Results:** a unit shock, roughly the average decade rise in exposure, lowers the share of women aged 18–39 currently married by 0.95 points and raises the never-married share by 1.2. Male-intensive shocks reduce marriage; female-intensive ones push the other way.

## In the Literature: Derenoncourt (2022)

**The question.** Did the Great Migration (1940–1970) lower the upward mobility northern cities offer children today? OLS fails: migrants went where jobs were growing, and booming cities differ in mobility for other reasons.

**The data.** 130 non-southern commuting zones.  $D$ : 1940–1970 rise in urban Black population over 1940 urban population.  $Y$ : adult income rank of children born 1980–1986, IRS records (Chetty and Hendren 2018).

**The instrument** (Boustan 2010, at the county level): with  $j$  a southern county, the share is  $\omega_{jc}$  = the fraction of county  $j$ 's 1935–1940 Black migrants who settled in city  $c$  (1940 complete-count census), and the shock is  $\hat{m}_j$  = Black out-migration from  $j$ , 1940–1970, *predicted* from county economies: cotton and tenant farming, tobacco, mining, and WWII defense spending:

$$Z_{CZ} = \frac{\sum_{c \in CZ} \sum_j \omega_{jc} \hat{m}_j}{\text{urban population}_{CZ, 1940}} \quad \sum_k s_{\ell k} g_k, k = 1, 200+ \text{ counties}$$

## Derenoncourt: the Shock

$\hat{m}_j$ : Black out-migration from southern county  $j$ , 1940–1970, predicted from the county's economy one census earlier. Predictors (Boustan 2010):

- **Cotton**: cotton's share of farm acreage; share of farms run by tenants. Cotton mechanized after WWII and displaced tenants and sharecroppers: a push out.
- **Farming and tobacco**: agriculture's share of the labor force; tobacco state; their interaction.
- **Mining**: mining's share of the labor force; mining state (Oklahoma, Texas); their interaction.
- **WWII defense spending** per capita: war plants and shipyards kept workers in the South.

**LASSO, then prediction.** Each decade, LASSO (5-fold cross-validation) picks which predictors forecast the county's net migration rate; OLS on those gives the predicted rate.  $\hat{m}_j = \text{predicted rate} \times \text{Black population}$ , summed over the 1940s, 1950s, and 1960s. All are conditions in southern counties, argued to be unrelated to later mobility in northern cities.

## Derenoncourt: the Three Conditions

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|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevance    | One percentile more predicted increase gives 0.297 (0.076) percentile more actual increase; $F = 15.3$ . <b>Testable, adequate, not strong.</b>                                                                                                                                                                                                                                                                                                                         |
| Independence | The shocks story (Borusyak, Hull, and Jaravel 2022): early migrants did not pick cities at random, but southern push shocks are unrelated to northern mobility. Placebo (her Table 3): $Z$ does not predict 1900–1940 school attendance of teens with low-status fathers (0.011 to 0.025, s.e. near 0.03) or 1940 education.                                                                                                                                            |
| Exclusion    | Southern shocks reach northern children only through who arrived. Threats: a shock hitting origin and destination together (Virginia's and Baltimore's defense plants), or another migration. Defenses: shocks residualized on southern state fixed effects and a birth-state version give the same answer; controls for white southern migration, European migration instrumented by railway access (Sequeira, Nunn, and Qian 2020), and a manufacturing Bartik shock. |

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## Derenoncourt: Main Results

Her Tables 4, 6, 7; 130 zones (129 for Black men).  $D$  enters as its percentile; one SD = 29 percentiles:

| outcome, parents at $p = 25$      | OLS               | first stage      | reduced form      | 2SLS              |
|-----------------------------------|-------------------|------------------|-------------------|-------------------|
| household income rank, all        | -0.066<br>(0.010) | 0.297<br>(0.076) | -0.037<br>(0.010) | -0.125<br>(0.033) |
| individual income rank, Black men | -0.065<br>(0.012) | 0.310<br>(0.074) | -0.026<br>(0.011) | -0.085<br>(0.033) |
| individual income rank, white men | -0.014<br>(0.010) | 0.297<br>(0.076) | -0.012<br>(0.009) | -0.041<br>(0.031) |

- **Reading.**  $-0.037/0.297 = -0.125$ : one SD more Great Migration, 3.6 rank points lower (mean 45.8).
- **Black versus white men**, same outcome, both born 1978–83:  $-0.085$  against  $-0.041$  (not significant), 2.5 against 1.2 rank points per SD. The penalty fell mainly on Black men.
- The Migration accounts for 27% of today's northern Black–white gap.

# Practice

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## A Reader's Checklist ★

Adapted from Sovey and Green (2011). Any IV paper should report:

1. The instrument, and an explicit argument for exclusion, not just a citation to a prior paper that used it.
2. The **first stage**, its coefficient, and its robust  $F$ .
3. The **reduced form** ( $Y$  on  $Z$ ), with its estimate and interval. It needs no instrument-strength assumption, so it is the most robust piece of evidence in the design, and if it is indistinguishable from zero the IV estimate rests on very little.
4. The **OLS** estimate alongside, so the reader can see the direction and size of the correction, and whether it goes the way theory predicts.
5. Who the **compliers** are, and their share.
6. Weak-instrument-robust inference if  $F$  is not comfortably large.
7. Placebo checks: outcomes and subgroups  $Z$  should not affect.

## A Reader's Checklist ★ (cont.)

If a paper reports only the 2SLS coefficient, that is a red flag.

## When IV Is the Wrong Tool ★

- **No credible instrument exists.** This is the normal state of affairs. An implausible instrument is worse than an honest selection-on-observables analysis with sensitivity bounds.
- **The LATE is not the quantity of interest.** If your policy question concerns always-takers, the compliers' effect does not answer it.
- **Interpretation drifts.** Papers routinely estimate a LATE and then discuss it as if it were an ATE.
- **Monotonicity is doing hidden work.** In judge-leniency and examiner designs, it must hold for *every* judge simultaneously, a much stronger claim than it first appears.

## When IV Is the Wrong Tool ★ (cont.)

### The design-based cousins

When the instrument is a discontinuity in an assignment rule, you have **RDD** (Week 9), and indeed fuzzy RD *is* local IV. When it is a policy change over time, you have **DID** (Week 10).

## Looking Ahead

- IV replaces “I measured all confounders” with “I found variation that is as good as random.” Both are assumptions; the second is easier to argue about substantively, and easier for readers to attack, which is a virtue.
- The price is the estimand: LATE, for compliers you cannot name, defined by the instrument you chose.
- Notice how much identification work was done by *functional form* in AJR, “linear in distance from the equator, plus continent dummies.” Week 13 asks briefly what can be done when you are unwilling to assume that.

### Midterm

**Thursday, October 15**, in class, covering Weeks 1–7. Open book and open note, timed, no coding. Tuesday October 13 is Fall break.

# References

- Abadie, Alberto. 2003. “Semiparametric Instrumental Variable Estimation of Treatment Response Models.” *Journal of Econometrics* 113(2): 231–263.
- Acemoglu, Daron, Simon Johnson, and James A. Robinson. 2001. “The Colonial Origins of Comparative Development.” *American Economic Review* 91(5): 1369–1401.
- Aizer, Anna and Joseph J. Doyle Jr. 2015. “Juvenile Incarceration, Human Capital, and Future Crime: Evidence from Randomly Assigned Judges.” *Quarterly Journal of Economics* 130(2): 759–803.
- Aksoy, Ozan and Diego Gambetta. 2022. “Commitment through Sacrifice: How Longer Ramadan Fasting Strengthens Religiosity and Political Islam.” *American Sociological Review* 87(4): 555–583.
- Anderson, T.W. and Herman Rubin. 1949. “Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations.” *Annals of Mathematical Statistics* 20(1): 46–63.
- Angrist, Joshua D. 1990. “Lifetime Earnings and the Vietnam Era Draft Lottery: Evidence from Social Security Administrative Records.” *American Economic Review* 80(3): 313–336.

## References (cont.)

- Angrist, Joshua D. and Alan B. Krueger. 1991. “Does Compulsory School Attendance Affect Schooling and Earnings?” *Quarterly Journal of Economics* 106(4): 979–1014.
- Angrist, Joshua D. and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics*. Princeton University Press. [Ch. 4]
- Angrist, Joshua D., Guido W. Imbens, and Alan B. Krueger. 1999. “Jackknife Instrumental Variables Estimation.” *Journal of Applied Econometrics* 14(1): 57–67.
- Angrist, Joshua D., Guido W. Imbens, and Donald B. Rubin. 1996. “Identification of Causal Effects Using Instrumental Variables.” *Journal of the American Statistical Association* 91(434): 444–455.
- Autor, David H., David Dorn, and Gordon H. Hanson. 2013. “The China Syndrome: Local Labor Market Effects of Import Competition in the United States.” *American Economic Review* 103(6): 2121–2168.

## References (cont.)

- Autor, David, David Dorn, and Gordon Hanson. 2019. “When Work Disappears: Manufacturing Decline and the Falling Marriage Market Value of Young Men.” *American Economic Review: Insights* 1(2): 161–178.
- Bartik, Timothy J. 1991. *Who Benefits from State and Local Economic Development Policies?* Kalamazoo: W.E. Upjohn Institute.
- Bhuller, Manudeep, Gordon B. Dahl, Katrine V. Løken, and Magne Mogstad. 2020. “Incarceration, Recidivism, and Employment.” *Journal of Political Economy* 128(4): 1269–1324.
- Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2022. “Quasi-Experimental Shift-Share Research Designs.” *Review of Economic Studies* 89(1): 181–213.
- Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2025. “A Practical Guide to Shift-Share Instruments.” *Journal of Economic Perspectives* 39(1): 181–204.

## References (cont.)

- Bound, John, David A. Jaeger, and Regina M. Baker. 1995. “Problems with Instrumental Variables Estimation When the Correlation Between the Instruments and the Endogenous Explanatory Variable Is Weak.” *Journal of the American Statistical Association* 90(430): 443–450.
- Campante, Filipe and David Yanagizawa-Drott. 2015. “Does Religion Affect Economic Growth and Happiness? Evidence from Ramadan.” *Quarterly Journal of Economics* 130(2): 615–658.
- Boustan, Leah Platt. 2010. “Was Postwar Suburbanization ‘White Flight’? Evidence from the Black Migration.” *Quarterly Journal of Economics* 125(1): 417–443.
- Card, David. 1995. “Using Geographic Variation in College Proximity to Estimate the Return to Schooling.” In *Aspects of Labour Market Behaviour: Essays in Honour of John Vanderkamp*, ed. L. N. Christofides, E. K. Grant, and R. Swidinsky. University of Toronto Press, 201–222.
- Chetty, Raj and Nathaniel Hendren. 2018. “The Impacts of Neighborhoods on Intergenerational Mobility II: County-Level Estimates.” *Quarterly Journal of Economics* 133(3): 1163–1228.

## References (cont.)

- Chyn, Eric, Brigham Frandsen, and Emily C. Leslie. 2025. “Examiner and Judge Designs in Economics: A Practitioner’s Guide.” *Journal of Economic Literature* 63(2): 401–439.
- de Vaan, Mathijs and Toby Stuart. 2019. “Does Intra-household Contagion Cause an Increase in Prescription Opioid Use?” *American Sociological Review* 84(4): 577–608.
- Derenoncourt, Ellora. 2022. “Can You Move to Opportunity? Evidence from the Great Migration.” *American Economic Review* 112(2): 369–408.
- Dufour, Jean-Marie. 1997. “Some Impossibility Theorems in Econometrics with Applications to Structural and Dynamic Models.” *Econometrica* 65(6): 1365–1387.
- de Chaisemartin, Clément. 2017. “Tolerating Defiance? Local Average Treatment Effects without Monotonicity.” *Quantitative Economics* 8(2): 367–396.
- Dobbie, Will, Jacob Goldin, and Crystal S. Yang. 2018. “The Effects of Pretrial Detention on Conviction, Future Crime, and Employment: Evidence from Randomly Assigned Judges.” *American Economic Review* 108(2): 201–240.

## References (cont.)

- Evans, William N. and Emily G. Owens. 2007. “COPS and Crime.” *Journal of Public Economics* 91(1–2): 181–201.
- Frandsen, Brigham, Lars Lefgren, and Emily Leslie. 2023. “Judging Judge Fixed Effects.” *American Economic Review* 113(1): 253–277.
- Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. 2020. “Bartik Instruments: What, When, Why, and How.” *American Economic Review* 110(8): 2586–2624.
- Green, Donald P. and Daniel Winik. 2010. “Using Random Judge Assignments to Estimate the Effects of Incarceration and Probation on Recidivism among Drug Offenders.” *Criminology* 48(2): 357–387.
- Harding, David J., Jeffrey D. Morenoff, Anh P. Nguyen, and Shawn D. Bushway. 2017. “Short- and Long-Term Effects of Imprisonment on Future Felony Convictions and Prison Admissions.” *Proceedings of the National Academy of Sciences* 114(42): 11103–11108.

## References (cont.)

- Harding, David J., Jeffrey D. Morenoff, Anh P. Nguyen, and Shawn D. Bushway. 2018. “Imprisonment and Labor Market Outcomes: Evidence from a Natural Experiment.” *American Journal of Sociology* 124(1): 49–110.
- Harding, David J., Jeffrey D. Morenoff, Anh P. Nguyen, Shawn D. Bushway, and Ingrid A. Binswanger. 2019. “A Natural Experiment Study of the Effects of Imprisonment on Violence in the Community.” *Nature Human Behaviour* 3(7): 671–677.
- Imbens, Guido W. and Joshua D. Angrist. 1994. “Identification and Estimation of Local Average Treatment Effects.” *Econometrica* 62(2): 467–475.
- Kitagawa, Toru. 2015. “A Test for Instrument Validity.” *Econometrica* 83(5): 2043–2063.
- Kling, Jeffrey R. 2006. “Incarceration Length, Employment, and Earnings.” *American Economic Review* 96(3): 863–876.
- Lee, David S., Justin McCrary, Marcelo J. Moreira, and Jack Porter. 2022. “Valid  $t$ -Ratio Inference for IV.” *American Economic Review* 112(10): 3260–3290.

## References (cont.)

- Levitt, Steven D. 2002. “Using Electoral Cycles in Police Hiring to Estimate the Effects of Police on Crime: Reply.” *American Economic Review* 92(4): 1244–1250.
- Loeffler, Charles E. 2013. “Does Imprisonment Alter the Life Course? Evidence on Crime and Employment from a Natural Experiment.” *Criminology* 51(1): 137–166.
- Mogstad, Magne, Alexander Torgovitsky, and Christopher R. Walters. 2021. “The Causal Interpretation of Two-Stage Least Squares with Multiple Instrumental Variables.” *American Economic Review* 111(11): 3663–3698.
- Morgan, Stephen L. and Christopher Winship. 2015. *Counterfactuals and Causal Inference*, 2nd ed. Cambridge University Press. [Ch. 9]
- Sequeira, Sandra, Nathan Nunn, and Nancy Qian. 2020. “Immigrants and the Making of America.” *Review of Economic Studies* 87(1): 382–419.
- Sharkey, Patrick and Gerard Torrats-Espinosa. 2017. “The Effect of Violent Crime on Economic Mobility.” *Journal of Urban Economics* 102: 22–33.

## References (cont.)

- Noack, Claudia. 2021. “Sensitivity of LATE Estimates to Violations of the Monotonicity Assumption.” arXiv:2106.06421.
- Sharkey, Patrick, Gerard Torrats-Espinoza, and Delaram Takyar. 2017. “Community and the Crime Decline: The Causal Effect of Local Nonprofits on Violent Crime.” *American Sociological Review* 82(6): 1214–1240.
- Sovey, Allison J. and Donald P. Green. 2011. “Instrumental Variables Estimation in Political Science: A Reader’s Guide.” *American Journal of Political Science* 55(1): 188–200.
- Staiger, Douglas and James H. Stock. 1997. “Instrumental Variables Regression with Weak Instruments.” *Econometrica* 65(3): 557–586.
- Stock, James H. and Motohiro Yogo. 2005. “Testing for Weak Instruments in Linear IV Regression.” In *Identification and Inference for Econometric Models*, ed. Andrews and Stock. Cambridge University Press, 80–108.
- Wilson, William Julius. 1996. *When Work Disappears: The World of the New Urban Poor*. New York: Knopf.